Scala debugging

Intermediate Level (10)

Q1

Find the error.

```
val x: Int = "10"
println(x + 5)
```

Ans : error: type mismatch

Q2

Identify issue.

```
var total = 0

for(i <- 1 to 5){
total = i
}

println(total)
```

Expected result should be **sum of numbers**.

Ans : here the variable total get reassigned in every iteration instead of adding the value

Q3

```
val list = List(1,2,3,4)

list.foreach(x =>
println(x)
println(x*2)
)
```

Why only first print works?

Ans : because in foreach loop only one argument is accepting

Q4

```
val numbers = List(1,2,3)

val result = numbers.map(x => x*2)

println(numbers)
```

Why result is not printed?

Ans : because instead of result the input list numbers is printing

Q5

```
def add(a:Int,b:Int)
{
a+b
}

println(add(5,10))
```

Why compiler error occurs?

Ans : because add is not returning any thing it is a procedure

Q6

```
val data = Map("a"->1,"b"->2)

println(data("c"))
```

Why runtime error occurs?

Ans : key not found error

Q7

```
var x = 5

if(x = 10){
println("Equal")
}
```

Find the mistake.

Ans : instead of ‘=’ use ‘==’ to check equal or not

Q8

val list = List(1,2,3)

list.map(x => println(x))

Why map is not ideal here?

Ans : we can directly print using foreach loop over a collection also map is used to transform the value here no transformation tacking place

Q9

def greet(name:String){
println("Hello "+name)
}

greet()

What is the error?

Ans : the function missing a required parameter name

Q10

val list = List(1,2,3)

for(i <- list)
println(i)
println("Done")

Why only one statement inside loop?

Ans : because to include more then one statement in the loop add { } to make it as a block

Q11

class Person(name:String){
println(name)
}

val p = new Person

Why compilation fails?

Ans : the primary constructor require the parameter name here it is missing

Q12

```
class User{
private var age:Int = 0
}

val u = new User
println(u.age)
```

Why error occurs?

Ans : age is a private variable so it can be accessed only through getters and setters

Q13

```
object Test{
def main(args:Array[String]){
println("Hello")
}
}
```

Why program may not run properly in Scala 3 environments?

Ans : because the main function missing return type

Q14

```
class A{
def show(){
println("A")
}
}

class B extends A{
def show(){
println("B")
}
}
```

What concept is used and what improvement needed?

Ans : the concepts used here is inheritance and method overriding and the improvement needed is usage of override keyword

Q15

```
object Config{
var url="localhost"
}

class Service{
val url="production"
println(Config.url)
}
```

Why incorrect value printed?

Ans : because the url associated with the singleton config is printing here